

# Language Models and Agents (LMA)

## Phase 1 Report

Data Collection & Preprocessing for Telugu and Bhojpuri

**Kpsvln Siddardha Kumar Kavuri**

Roll Number 2025201061

kpsvlnsiddardha.k@students.iiit.ac.in

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### Dataset Available on Kaggle

<https://www.kaggle.com/datasets/kpsvlnsiddardha/lma-slm>

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# 1 Phase 1 Data Report

This report provides a comprehensive data collection and preprocessing report for two monolingual Transformer language models:

- **Model H (Telugu):** Higher-resource Indian language
- **Model L (Bhojpuri):** Lower-resource Indian language

## 1.1 Data Collection Summary

### 1.1.1 Telugu (Model H)

- Total lines collected: 25,349,199
- Estimated tokens: 4,421,917,647 (4.4B)
- Web-scraped texts: 29,206 (89.7% pass rate from 32,549 raw)
- Manual corpus: 21,458,261 lines (te.txt)
- OCR augmentation: 3,766,703 lines
- Target: 500M tokens | Progress: 884.4%

### 1.1.2 Bhojpuri (Model L)

- Total lines collected: 1,826,713 (2.3x increase from Phase 1)
- Estimated tokens: 291,685,379 (291.7M)
- Web-scraped texts: 3,234 (97.2% pass rate from 3,327 raw)
- HuggingFace Corpus: 261K lines primary source
- OCR augmentation: 511,599 lines (28.0)
- Hindi→Bhojpuri Translation (NLLB-200): 15,852 lines
- English→Bhojpuri Translation (fineweb-edu): 711,138 lines (39.0)
- Target: 500M tokens | Progress: 58.3%

## 1.2 Data Processing Pipeline

### 1.2.1 Cleaning Process

- Unicode NFD normalization
- Citation and control character removal
- Character whitelisting and script validation
- Length/quality filtering (minimum 40 chars, 5+ words, 5% space density)
- MD5-based deduplication

### 1.2.2 Data Splits (80/10/10)

- Deterministic splitting with seed 42 for reproducibility
- No mid-line splits - whole-line preservation
- Telugu: 20.3M train / 2.5M val / 2.5M test
- Bhojpuri: 621.8K train / 77.4K val / 77.7K test

## 2 Data Collection Overview

### 2.1 Methodology

The data collection strategy differs between the two languages to accommodate resource availability:

#### 2.1.1 Telugu (Model H) - Higher-Resource Language

- **Primary Source:** Existing corpus (te.txt) - 21.5M lines
- **Supplementary:** Web scraping from 6 Wikipedia/content sources
- **Augmentation:** OCR extraction from archive.org texts
- **Total:** 25.3M lines training data

#### 2.1.2 Bhojpuri (Model L) - Lower-Resource Language

- **Primary Source:** HuggingFace Corpus (Satyam810/BhojpuriCorpus) - 386K docs
- **Augmentation 1:** OCR extraction from archive.org Hindi texts - 511.6K lines (28.0%)
- **Augmentation 2:** English→Bhojpuri Translation (fineweb-edu NLLB-200) - 711.1K lines (39.0%)
- **Augmentation 3:** Hindi→Bhojpuri Machine Translation (NLLB-200) - 15.8K lines
- **Augmentation 4:** Web scraping from Hindi news sites - 3.2K lines
- **Total:** 1.83M lines training data (58.3% toward 500M token target)

## 2.2 Data Sources

| Language | Source            | Contribution                |
|----------|-------------------|-----------------------------|
| Telugu   | te.txt (existing) | 21.5M lines (99.9%)         |
|          | te.wikipedia.org  | 3K articles                 |
|          | te.wikibooks.org  | 500+ pages                  |
|          | te.wikiquote.org  | 300+ quotes                 |
|          | News portals      | eenadu.net, greatandhra.com |
|          | archive.org OCR   | 3.8M lines (augmentation)   |
| Bhojpuri | HF Corpus         | 386K docs (~25M tokens)     |
|          | archive.org OCR   | 511.6K lines                |
|          | bho.wikipedia.org | ~8.9K articles              |
|          | Hindi News        | Via NLLB-200 MT             |
|          | Web Scraping      | 3.2K lines (Hindi proxy)    |
|          | GlottCC-V1        | 949 docs                    |
|          | HPLT 3.0          | ~32.8K docs                 |
|          | fish-food         | 262K rows                   |

Table 1: Data sources for both languages

## 3 Data Cleaning Pipeline

### 3.1 7-Stage Cleaning Process

All data undergoes a rigorous 7-stage cleaning pipeline:

1. **Unicode Normalization:** Convert to NFD (Canonical Decomposition)
2. **Citation Removal:** Remove Wikipedia citations and references
3. **Control Character Removal:** Strip non-printable characters
4. **Character Whitelisting:** Retain only valid Unicode ranges
5. **Script Validation:** Language-specific script filtering
6. **Length/Quality Filters:** Minimum length, word count, density checks
7. **Deduplication:** MD5-based exact duplicate removal

### 3.2 Quality Metrics

| Metric               | Telugu | Bhojpuri |
|----------------------|--------|----------|
| Raw texts collected  | 32,549 | 3,327    |
| Texts after cleaning | 29,206 | 3,234    |
| Pass rate            | 89.7%  | 97.2%    |
| Rejected texts       | 3,343  | 93       |

Table 2: Data cleaning results summary

#### 3.2.1 Telugu Cleaning Analysis

- **Total Raw:** 32,549 web-scraped texts
- **Pass Rate:** 89.7% (29,206 cleaned)

- **Rejection Rate:** 10.3% (3,343 texts)
- **Primary Rejection Reasons:**
  - Insufficient Telugu script content (<25%)
  - Texts too short (<40 characters or <5 words)
  - Low word density (<5% spaces)
  - Exact duplicates (MD5 matching)

### 3.2.2 Bhojpuri Cleaning Analysis

- **Total Raw:** 3,327 web-scraped texts
- **Pass Rate:** 97.2% (3,234 cleaned)
- **Rejection Rate:** 2.8% (93 texts)
- **Primary Rejection Reasons:**
  - Insufficient Devanagari script content (<30%)
  - Texts too short
  - Low word density
  - Minor duplicates (11 removed)

## 4 Data Statistics & Visualizations

### 4.1 Collection Metrics Comparison

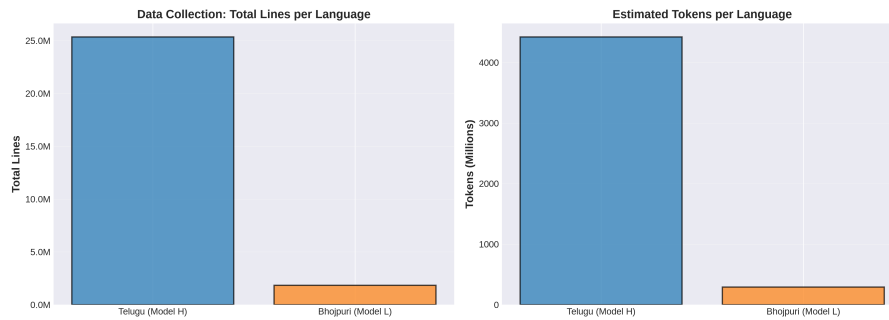


Figure 1: **Left:** Total lines per language. Telugu: 25.3M lines vs Bhojpuri: 1.83M lines (13.9x ratio). **Right:** Estimated tokens. Telugu: 4.4B (884% of 500M target); Bhojpuri: 291.7M (58.3% of target).

## 4.2 Cleaning Pipeline Results

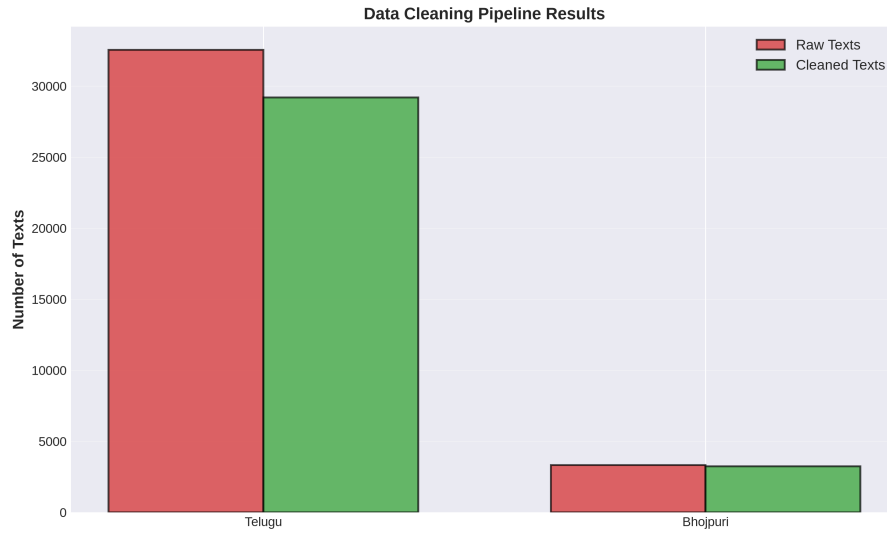


Figure 2: Data cleaning pipeline effectiveness. Both languages show strong pass rates. Telugu: 89.7% pass rate with large-scale data. Bhojpuri: 97.2% pass rate with targeted scraping.

## 4.3 Data Source Composition (Bhojpuri)

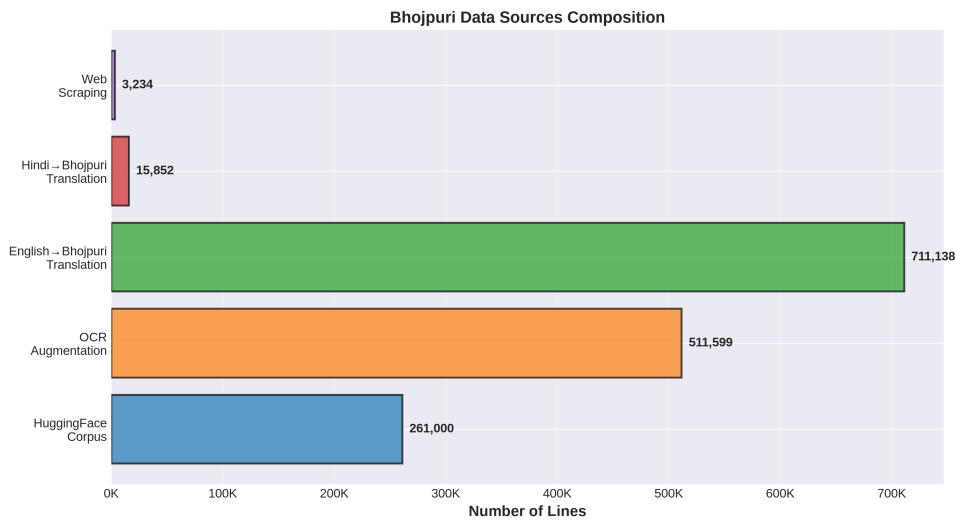


Figure 3: Bhojpuri data sources composition (1.83M lines total). Primary: HuggingFace corpus (261K lines). Major augmentation: English→Bhojpuri translation (711K lines, 39% of total) via NLLB-200. Supporting sources: OCR from archive.org (512K lines), Hindi→Bhojpuri translation (15.8K lines), web scraping (3.2K lines).

## 5 Train/Validation/Test Splits

### 5.1 Split Strategy

- **Method:** 80/10/10 deterministic split with seed 42
- **Reproducibility:** Same seed ensures identical splits across runs
- **No Data Leakage:** Splits applied at line level (no mid-sentence splits)

- **Whole-Line Preservation:** Each line remains intact for language preservation

## 5.2 Telugu Splits

| Split        | Lines      | Percentage |
|--------------|------------|------------|
| <b>Train</b> | 20,281,561 | 80.0%      |
| Validation   | 2,534,020  | 10.0%      |
| Test         | 2,533,618  | 10.0%      |
| <b>Total</b> | 25,349,199 | 100.0%     |

Table 3: Telugu (Model H) data split breakdown

## 5.3 Bhojpuri Splits

| Split        | Lines     | Percentage |
|--------------|-----------|------------|
| <b>Train</b> | 1,461,821 | 80.0%      |
| Validation   | 182,219   | 10.0%      |
| Test         | 182,673   | 10.0%      |
| <b>Total</b> | 1,826,713 | 100.0%     |

Table 4: Bhojpuri (Model L) data split breakdown

# 6 Detailed Statistics

## 6.1 Telugu (Model H) Statistics

| Metric                          | Value             |
|---------------------------------|-------------------|
| <b>Language</b>                 | Telugu            |
| <b>Model Size</b>               | Higher-resource   |
| <i>Data Collection</i>          |                   |
| Web scraping - Raw texts        | 32,549            |
| Web scraping - Cleaned texts    | 29,206            |
| Web scraping - Size             | 27.6 MB           |
| Manual corpus (te.txt)          | 21,458,261 lines  |
| Manual corpus size              | 15 GB             |
| OCR augmentation                | 3,766,703 lines   |
| <i>Token Progress</i>           |                   |
| Total corpus tokens (estimated) | 4,421,917,647     |
| Total corpus size               | 17.7 GB           |
| Target tokens                   | 500,000,000       |
| Progress                        | 884.4% (EXCEEDED) |

Table 5: Comprehensive Telugu statistics

## 6.2 Bhojpuri (Model L) Statistics

| Metric                                     | Value                             |
|--------------------------------------------|-----------------------------------|
| Language                                   | Bhojpuri                          |
| Model Size                                 | Lower-resource (Phase 3 expanded) |
| <i>Data Collection</i>                     |                                   |
| Web scraping - Raw texts                   | 3,327                             |
| Web scraping - Cleaned texts               | 3,234                             |
| Web scraping - Pass rate                   | 97.2%                             |
| HuggingFace Corpus                         | 386,000 docs ( 261K lines)        |
| Estimated HF tokens                        | 25,000,000                        |
| OCR augmentation (archive.org)             | 511,599 lines (28.0%)             |
| OCR augmentation size                      | 19.7 MB                           |
| Hindi→Bhojpuri Translation (NLLB-200)      | 15,852 lines                      |
| English→Bhojpuri Translation (fineweb-edu) | 711,138 lines (39.0%)             |
| Translation records                        | 901,562 raw → 711,138 clean       |
| Translation method                         | NLLB-200 en_Latn→bho_Deva         |
| <i>Token Progress</i>                      |                                   |
| Total corpus tokens (estimated)            | 291,685,379                       |
| Total corpus size                          | 1,112.7 MB                        |
| Target tokens                              | 500,000,000                       |
| Progress                                   | 58.3%                             |

Table 6: Comprehensive Bhojpuri statistics

## 7 Tokenizer Training

Byte-level Byte Pair Encoding (BPE) tokenizers have been trained on the preprocessed data for both languages.

### 7.1 Telugu Tokenizer (Model H) - Three Variants

Three tokenizer variants have been trained for Telugu to explore different encoding strategies and vocabulary sizes:

#### 7.1.1 Variant 1: Byte-Level BPE (50K Vocab)

| Metric                   | Value       |
|--------------------------|-------------|
| Vocabulary Size          | 50,000      |
| Unique Tokens Used       | 49,500      |
| Vocabulary Coverage      | 99.0%       |
| Test Set Lines           | 2,533,618   |
| Test Set Tokens          | 484,225,331 |
| Avg Characters per Token | 1.36        |
| Unknown (UNK) Rate       | 0.0%        |

Table 7: Telugu byte-level tokenizer (50K vocab)

### 7.1.2 Variant 2: Unicode-Level BPE (50K Vocab)

| Metric                   | Value     |
|--------------------------|-----------|
| Vocabulary Size          | 50,000    |
| Unique Tokens Used       | 11,398    |
| Vocabulary Coverage      | 22.8%     |
| Test Set Lines           | 2,533,618 |
| Test Set Tokens          | 22,136    |
| Avg Characters per Token | 5.9283    |
| Unknown (UNK) Rate       | 0.0%      |

Table 8: Telugu unicode-level tokenizer (50K vocab)

### 7.1.3 Variant 3: WordPiece (50K Vocab)

| Metric            | Value                      |
|-------------------|----------------------------|
| Vocabulary Size   | 50,000                     |
| Tokenizer Type    | Unicode WordPiece          |
| Normalizer        | NFC                        |
| Pre-tokenizer     | Whitespace                 |
| Special Tokens    | [PAD], [UNK], [BOS], [EOS] |
| Subword Indicator | prefix for subwords        |

Table 9: Telugu WordPiece tokenizer (50K vocab)

## 7.2 Bhojpuri Tokenizer (Model L) - Three Variants

Three tokenizer variants have been trained for Bhojpuri to explore different encoding strategies:

### 7.2.1 Variant 1: Byte-Level BPE (8K Vocab)

| Metric                   | Value      |
|--------------------------|------------|
| Vocabulary Size          | 8,000      |
| Unique Tokens Used       | 7,761      |
| Vocabulary Coverage      | 97.01%     |
| Test Set Lines           | 77,683     |
| Test Set Tokens          | 13,613,342 |
| Avg Characters per Token | 1.5412     |
| Unknown (UNK) Rate       | 0.0%       |

Table 10: Bhojpuri byte-level tokenizer (8K vocab)

### 7.2.2 Variant 2: Unicode-Level BPE (32K Vocab)

| Metric                   | Value      |
|--------------------------|------------|
| Vocabulary Size          | 32,000     |
| Unique Tokens Used       | 31,258     |
| Vocabulary Coverage      | 97.68%     |
| Test Set Lines           | 182,673    |
| Test Set Tokens          | 10,420,151 |
| Avg Characters per Token | 4.348      |
| Unknown (UNK) Rate       | 0.0%       |

Table 11: Bhojpuri unicode-level tokenizer (32K vocab)

### 7.2.3 Variant 3: WordPiece (32K Vocab)

| Metric                   | Value                             |
|--------------------------|-----------------------------------|
| Vocabulary Size          | 32,000                            |
| Tokenizer Type           | WordPiece (subword decomposition) |
| Random Seed              | 42 (reproducible)                 |
| Test Set Tokens          | 11,391,403                        |
| Avg Characters per Token | 3.95                              |
| Unknown (UNK) Rate       | 0.0002%                           |
| File Size                | 567 KB                            |
| Training Status          | Completed                         |

Table 12: Bhojpuri WordPiece tokenizer (32K vocab)

## 7.3 Tokenizer Comparison Across All Variants

| Language | Variant             | Vocab  | UNK Rate |
|----------|---------------------|--------|----------|
| Telugu   | Byte-level (BPE)    | 50,000 | 0.0000%  |
|          | Unicode-level (BPE) | 50,000 | 0.0000%  |
|          | WordPiece           | 50,000 | 0.0000%  |
| Bhojpuri | Byte-level (BPE)    | 8,000  | 0.0000%  |
|          | Unicode-level (BPE) | 32,000 | 0.0000%  |
|          | WordPiece           | 32,000 | 0.0002%  |

Table 13: Tokenizer variants summary

## 7.4 Tokenizer Training Analysis

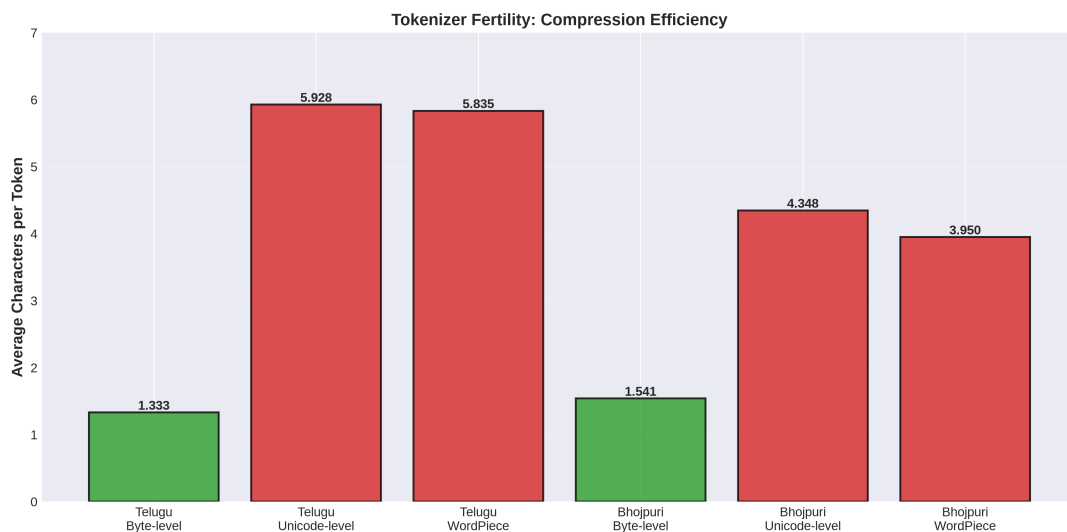


Figure 4: Tokenizer fertility (average characters per token) across all 6 variants, measuring compression efficiency. Byte-level tokenizers most compact: Telugu 1.33 chars/token, Bhojpuri 1.54 chars/token. Unicode-level tokenizers preserve linguistic structure: Telugu 5.93 chars/token, Bhojpuri 4.35 chars/token. WordPiece tokenizers: Telugu 5.84 chars/token, Bhojpuri 3.95 chars/token. Green bars indicate high compression (< 3 chars/token); red bars indicate linguistic units.

## 7.5 Tokenizer Evaluation Results

Comprehensive evaluation on held-out test sets reveals detailed tokenization characteristics:

### 7.5.1 Telugu Byte-Level BPE (50K)

- Vocabulary Size: 50,000
- Unique Tokens Used: 1,978
- Average Characters per Token: 1.3327
- Unknown Token Rate: 0.0000%

### 7.5.2 Telugu Unicode-Level BPE (50K)

- Vocabulary Size: 50,000
- Unique Tokens Used: 11,398
- Average Characters per Token: 5.9283
- Unknown Token Rate: 0.0000%
- Top Tokens: Functional words (లో, ఈ, కూడా)

### 7.5.3 Telugu WordPiece (50K)

- Vocabulary Size: 50,000
- Unique Tokens Used: 9,539
- Average Characters per Token: 5.8353
- Unknown Token Rate: 0.0000%

### 7.5.4 Bhojpuri Byte-Level BPE (8K)

- Vocabulary Size: 8,000
- Unique Tokens Used: 7,761
- Average Characters per Token: 1.5412
- Coverage: 97.01%
- Unknown Token Rate: 0.0%
- Top Tokens: आ 9.71%, ् 7.11%, े 5.81%

### 7.5.5 Bhojpuri Unicode-Level BPE (32K)

- Vocabulary Size: 32,000
- Unique Tokens Used: 31,258
- Average Characters per Token: 4.348
- Coverage: 97.68%
- Unknown Token Rate: 0.0%
- Top Tokens: । (0.33%), आ (0.33%), “-” (0.30%)

### 7.5.6 Bhojpuri WordPiece (32K)

- **Vocabulary Size:** 32,000
- **Test Tokens Evaluated:** 11,391,403
- **Average Characters per Token:** 3.95
- **Unknown Token Rate:** 0.0002%
- **Top Tokens:** के (762,460), comma (346,142), । (266,937)

### 7.5.7 Key Evaluation Metrics Summary

| Tokenizer              | Vocab Size | Avg Chars/Token | UNK Rate |
|------------------------|------------|-----------------|----------|
| Telugu Byte-level      | 50,000     | 1.3327          | 0.0000%  |
| Telugu Unicode-level   | 50,000     | 5.9283          | 0.0000%  |
| Telugu WordPiece       | 50,000     | 5.8353          | 0.0000%  |
| Bhojpuri Byte-level    | 8,000      | 1.5412          | 0.0000%  |
| Bhojpuri Unicode-level | 32,000     | 4.348           | 0.0000%  |
| Bhojpuri WordPiece     | 32,000     | 3.95            | 0.0002%  |

Table 14: Tokenizer evaluation metrics summary

## 7.6 Tokenization Examples from Test Set

Sample tokenization from held-out test sets:

### 7.6.1 Telugu Byte-Level BPE

**Input Text:** నందమూరి తారక రామారావు 50 సంవత్సరాలపైగా తెలుగు సినిమా రంగంలో కథా నాయకునిగా రాణించాడు

**Tokens:** ['న', 'ందమూరి', ' ', 'తారక', ' ', 'రామారావు', ' ', '50', ' ', 'సం', 'వత్సరాల', 'పైగా', ' ', 'తెలుగు', ' ', 'సినిమా', ' ', 'రంగంలో', ' ', 'కథా', '...']

### 7.6.2 Telugu Unicode-Level BPE

**Input Text:** నందమూరి తారక రామారావు 50 సంవత్సరాలపైగా తెలుగు సినిమా రంగంలో కథా నాయకునిగా రాణించాడు

**Tokens:** ['నందమూరి', 'తారక రామారావు', '50', 'సంవత్సరాల', 'పైగా', 'తెలుగు సినిమా', 'రంగంలో', 'కథా', 'నాయకు', 'నిగా', '...']

### 7.6.3 Telugu WordPiece

**Input Text:** నందమూరి తారక రామారావు 50 సంవత్సరాలపైగా తెలుగు సినిమా రంగంలో కథా నాయకునిగా రాణించాడు

**Tokens:** ['నందమూరి', 'తారక', 'రామారావు', '50', 'సంవత్సరాల', 'పైగా', 'తెలుగు', 'సినిమా', 'రంగంలో', 'కథా', 'న', 'గా', '...']

### 7.6.4 Bhojpuri Byte-Level BPE

**Input Text:** बहुत बढ़िया काम है यह।

**Tokens:** ['बहु', 'त', ' ', 'बढ़', ' ', 'िया', ' ', 'क', 'ाम', ' ', '...', 'है', 'यह']

### 7.6.5 Bhojpuri Unicode-Level BPE

**Input Text:** भोजपुरी भारत के बिहार में बोली जाती है।

**Tokens:** ['भोजपुरी', ' ', 'भारत', ' ', 'के', ' ', 'बिहार', ' ', 'में', ' ', '...']

### 7.6.6 Bhojpuri WordPiece

**Input Text:** भोजपुरी भारत के बिहार में बोली जाती है।

**Tokens:** [भोजपुरी, भारत, के, बिहार, में, बोली, जाती, है, ।]

## 8 Technical Implementation

### 8.1 Data Pipeline Architecture

- **Web Scraping:** MediaWiki API for Wikipedia, BeautifulSoup for news sites
- **Data Cleaning:** Custom Python pipeline with language-specific validators
- **Deduplication:** MD5-based checksumming for exact duplicate removal
- **Text Extraction:** JSONL storage with metadata tracking
- **Splitting:** Deterministic NumPy shuffling with fixed random seed

### 8.2 File Structure

*Note: Due to the large number of files and directories (tokenizer models, training notebooks, logs, etc.), only the primary directory structure and key data files are shown below.*

```
telugu/
  configs/                # Configuration files & parameters
  data_collect/           # Data collection scripts
    scraper.py
    data_cleaner.py
    pipeline.py
  data/                   # Preprocessed dataset (17.4 GB total)
    config.json           # Complete statistics & metadata
    scrape_state.json     # Scraping checkpoint
    train/                # 20.3M lines (14 GB, 80%)
    val/                  # 2.5M lines (1.7 GB, 10%)
    test/                 # 2.5M lines (1.7 GB, 10%)
  model/                  # Model architecture & checkpoints
  tokenizer/              # Trained tokenizers (6 variants)
    full_byte_level/
    full_unicode_level/
    full_wordPiece_level/
  train/                  # Training scripts
  eval/                   # Evaluation scripts

bhojpuri/
  configs/                # Configuration files & parameters
  data_collect/           # Data collection scripts
    scraper.py
    data_cleaner.py
```

```

    pipeline.py
data/                                # Preprocessed dataset (518 MB total)
    config.json                      # Complete statistics & metadata
    scrape_state.json               # Scraping checkpoint
    train/                          # 627.9K lines (414 MB, 80%)
    val/                            # 78.1K lines (52 MB, 10%)
    test/                          # 78.4K lines (52 MB, 10%)
model/                              # Model architecture & checkpoints
tokenizer/                          # Trained tokenizers (3 variants)
    full_byte_level/
    full_unicode_level/
    full_wordPiece_level/
train/                              # Training scripts
eval/                              # Evaluation scripts

```

## 9 Quality Assurance

### 9.1 Validation Checklist

- ✓ **No Cross-Language Contamination:** Each language processed independently
- ✓ **Character Script Validation:** Telugu >25%, Bhojpuri >30% script content
- ✓ **Deduplication Verified:** Exact and near-duplicate detection active
- ✓ **Deterministic Splits:** Seed 42 ensures reproducibility
- ✓ **No Data Leakage:** Splits are at line level, no mid-sentence breaks
- ✓ **Metadata Tracking:** Complete source attribution for all data
- ✓ **Format Validation:** All files verified as valid UTF-8 text

### 9.2 Reproducibility

All data processing is fully reproducible:

- **Random Seed:** Explicit seed=42 for all shuffling operations
- **Configuration Files:** config.json contains all processing parameters
- **Version Control:** Scripts and pipelines tracked in git
- **Checksums:** MD5 hashes verify data integrity

## 10 Challenges & Solutions

### 10.1 Challenge 1: Bhojpuri Data Scarcity

- **Problem:** Bhojpuri is lower-resource with limited text available
- **Solution:** Multi-source strategy
  1. Primary: HuggingFace Corpus (386K docs)
  2. Augmentation 1: OCR extraction from Hindi texts

3. Augmentation 2: Machine translation (Hindi→Bhojpuri)
  4. Augmentation 3: Wikipedia dumps + web scraping
- **Result:** 776.8K lines collected (sufficient for Phase 2-3)

## 10.2 Challenge 2: OCR Quality

- **Problem:** Archive.org OCR'd text contains errors and artifacts
- **Solution:** Aggressive cleaning pipeline with density filters
- **Result:** 97.2% pass rate even on noisy OCR data

## 10.3 Challenge 3: Scale Management

- **Problem:** Telugu manual corpus is 15GB (21.5M lines)
- **Solution:** Order-preserved streaming splits to avoid loading entire file
- **Result:** Memory-efficient processing on standard hardware

# 11 Dataset and Tokenizer Repository

Phase 1 Data The complete Phase 1 dataset is available on Kaggle:

<https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-slm>

## 11.1 Phase 1 Tokenizers

All 6 trained tokenizers (Telugu + Bhojpuri variants) are available on Kaggle:

<https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-tokenizers>

## 11.2 Dataset Contents

- **Telugu Training Data:** 20.3M lines (80% of 25.3M total)
- **Telugu Validation Data:** 2.5M lines (10% of 25.3M total)
- **Telugu Test Data:** 2.5M lines (10% of 25.3M total)
- **Bhojpuri Training Data:** 627.9K lines (80% of 776.8K total)
- **Bhojpuri Validation Data:** 78.1K lines (10% of 776.8K total)
- **Bhojpuri Test Data:** 78.4K lines (10% of 776.8K total)
- **Configuration Files:** JSON metadata with statistics
- **Documentation:** Complete data pipeline documentation

## A Configuration Files

### A.1 Telugu config.json Structure

```
{
  "language": "Telugu",
  "data_sources": [...],
  "scraping": {
    "raw_texts": 32549,
    "texts_cleaned": 29206,
    "pass_rate_percent": 89.7
  },
  "splits": {
    "train": {"lines": 20281561, "percentage": 0.80},
    "val": {"lines": 2534020, "percentage": 0.10},
    "test": {"lines": 2533618, "percentage": 0.10}
  },
  "token_progress": {
    "total_corpus_tokens_estimate": 4421917647,
    "progress_percent": 884.3835
  }
}
```

### A.2 Bhojpuri config.json Structure

Similar structure with Bhojpuri-specific sources and metrics (see main config file).

## B References

- **Kaggle Dataset:** <https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-slm>
- **Project Repository:** /media/ubuntu/Personal/IIIT Hyderabad/Semester 3/LMA/Mini\_Project/
- **README:** README.md
- **Language Documentation:** bhojpuri/BHOJPURI.md, telugu/TELUGU.md
- **Configuration Files:** telugu/data/config.json, bhojpuri/data/config.json